

Chapter 9

Programming Project 7)

```
int power(int n, int pow)
{
    int result;

    if(pow == 0)
    {
        result = 1;
    }
    else if(pow % 2 == 0)
    {
        result = power(n, pow/ 2);
        result *= result;
    }
    else
    {
        result = n * power(n, --pow);
    }

    return result;
}
```

Chapter 10

1)

a) The f function

```
int a;
```

```
void f (int b)
```

```
{
```

```
    int c;
```

```
}
```

- Parameter: int b
- Variable: int c, a

b) The g function

```
int a;
```

```
void g(void)
```

```
{
```

```
    int d;
```

```
    {
```

```
        int e;
```

```
    }
```

```
}
```

- Parameter: void (no parameters)
- Variable: int d, e, a

c) The block in which e is declared

```
int a;  
  
void g(void)  
{  
    int d;  
    {  
        int e;  
    }  
}
```

- Function: void g
- Parameter: void (No parameters)
- Variables: int e, a, d
- It is only within the scope of the {} after int d

d) The main function

```
int a  
  
int main(void)  
{  
    int f;  
}
```

- Parameter: void (No parameters)
- Variable: int f, a

2)

a) The f function

```
int b, c;
```

```
void f (void)
```

```
{
```

```
    int b, d;
```

```
}
```

- Parameter: void (No parameters)
- Variable: int b, d, c

b) The g function

```
int b, c;
```

```
void g (int a)
```

```
{
```

```
    int c;
```

```
    {
```

```
        int a, d;
```

```
    }
```

```
}
```

- Parameter: int a
- Variable: int c, a, b, d

c) The block in which a and d are declared

```
int b, c;  
  
void g (int a)  
{  
    int c;  
    {  
        int a, d  
    }  
}
```

- Variable: int a, d, b, c
- Function: void g
- Parameter: int a
- Only within the scope of the {} after int c

d) The main function

```
int b, c;  
  
int main(void)  
{  
    int c, d;  
}
```

- Parameter: void (No parameters)
- Variable: int c, d, b, c

3)

If a program has only one function (main), it can only have one declaration of the type int i. If there were other functions, then those functions could also declare the type int i.